8th Grade Unit 8 Assessment Guide

General Information	Fluency of operations with rational numbers was an important outcome of grade 7. In terms of assessing the benchmarks on this assessment, it is assumed that students are fluent with rational numbers so theoretically students will achieve similarly with or without a calculator. This may not be the reality for many students. Teachers should consider how to balance the expectations of prior knowledge with the student's ability to show understanding of the MA.8.AR.2.1. If a calculator is not provided for this assessment, then teachers should consider how to monitor simple calculation errors. If a calculator is not provided, then the assessment may be too lengthy for some students. If teachers want to consider student fluency of rational numbers, then if a student makes a simple calculation error that is carried through their work, then the students should receive partial credit. Students may inadvertently write the wrong variable for an answer. This is a common mistake and should receive a minimal point deduction. It is suggested that if a student does it more than twice, 1 point is deducted for each occurrence past two. If teachers find that their students are struggling with rational numbers, then it may be helpful for the teacher to provide students with remediation. Teachers may choose to use data to help ascertain what students struggle with.
Calculator Information	The questions on the unit assessment are calculator neutral which means that a calculator does not give a student an advantage or disadvantage when showing their knowledge of the intended benchmark. Teachers should determine if the assessment is given with or without a calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

<table>
<tr><th>Question</th><th>1</th><th>Benchmark(s)</th><td>MA.8.AR.2.1</td></tr>
<tr><td colspan="2" rowspan="2">$$-35+10y+12y+3=12$$
$$22y-32=12$$
$$22y=44$$
$$y=2$$</td><th>Recommended Scoring

(8 Total Points)</th><td>Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 2 points for each algebraic error.</td></tr>
<tr><th>Additional Notes</th><td>If students are allowed to use a calculator, then consider what, if any, rounding is expected from your students. Without a calculator, students may provide $-\frac{26}{22}$ or $-\frac{13}{12}$ as solutions. If students have a calculator, then the answer is $-1.\overline{18}$. If students are not provided direction they may inadvertently round.</td></tr>
</table>

<table>
<tr><th>Question</th><th>2</th><th>Benchmark(s)</th><th>MA.8.AR.2.1</th></tr>
<tr><td colspan="2" rowspan="2">$$85.4 - 18.3h = -7.9h + 59.4$$
$$26 = 10.4h$$
$$2.5 = h$$</td><td>Recommended Scoring

(8 Total Points)</td><td>Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 2 points for each algebraic error.</td></tr>
<tr><td>Additional Notes</td><td>Students may provide $\frac{-26}{-10.4}$ as a solution. Setting expectations on solutions is left to the teacher. Teachers may want to consider that on a statewide assessment, $\frac{-26}{-10.4}$ would be an acceptable answer for Free Response questions but Multiple-Choice questions would have students select their answer from a list of values that are written in what would be simplest form or as a decimal.</td></tr>
</table>

Question	3	Benchmark(s)	MA.8.AR.2.1
$$-\frac{14}{5}+\frac{56}{5}k=\frac{6}{5}k+\frac{66}{5}$$ $$\frac{50}{5}k=\frac{80}{5}$$ $$10k=16$$ $$k=1.6$$		**Recommended Scoring** **(8 Total Points)**	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark. so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 2 points for each algebraic error.
		Additional Notes	Students may provide $\frac{16}{10}$ as a solution. Setting expectations on solutions is left to the teacher. Teachers may want to consider that on a statewide assessment, $\frac{16}{10}$ would be an acceptable answer for Free Response questions but Multiple-Choice questions would have students select their answer from a list of values that are written in what would be simplest form or as a decimal.

<table>
<tr><th>Question</th><th>4</th><th>Benchmark(s)</th><th>MA.8.AR.2.1</th></tr>
<tr><td colspan="2" rowspan="2">$$24r + 18 = 16 + 24r + 2$$
$$24r + 18 = 24r + 18$$
infinite solutions</td><td>Recommended Scoring

(8 Total Points)</td><td>Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 2 points for each algebraic error. Consider deducting 4 points for a student who has the correct work but gives something other than infinite solutions as the answer.</td></tr>
<tr><td>Additional Notes</td><td>Students who notice the answer is infinite solutions at an early step should not be punished for this recognition.</td></tr>
</table>

Question	5	Benchmark(s)	MA.8.AR.2.1
$\frac{2}{3}x - 18 + 4\frac{1}{3}x = -7$ $5x = 11$ $x = \frac{11}{5}$		**Recommended Scoring** **(8 Total Points)**	Students may show their work in a variety of ways. Consider reviewing work for minor errors that students carry through the problem. Students who make such errors may be showing their knowledge of the benchmark, so partial credit should be considered. It is suggested that minor errors have no more than a 2-point deduction. Consider deducting 2 points for each algebraic error.
		Additional Notes	Setting expectations on solutions is left to the teacher. Students may provide the answer as a decimal or as a mixed number.

<table>
<tr><td>Question</td><td>6</td><td>Benchmark(s)</td><td>MA.8.AR.4.2</td></tr>
<tr><td colspan="2">Answers will vary. Any line except for the original line, with a slope of -4 is acceptable.</td><td>Recommended Scoring

(4 Total Points)</td><td>No partial credit is suggested.</td></tr>
<tr><td>Question</td><td>7a</td><td>Benchmark(s)</td><td>MA.8.AR.4.1</td></tr>
<tr><td colspan="2">$5 = -\frac{9}{4}(8) + 2$
$5 = -18 + 2$
$5 = -16$

$5 = 3(8) - 19$
$5 = 24 - 19$
$5 = 5$
☐ No</td><td>Recommended Scoring

(6 Total Points)</td><td>Students may only show their work for $y = -\frac{9}{4}x + 2$ because it is false for the point $(8,5)$. Students who do this are showing an understanding that the solution must be true for both equations. On the other hand, if students only show their work for $y = 3x - 19$ and choose "yes," they are showing a lack of understanding.</td></tr>
<tr><td>Question</td><td>7b</td><td>Benchmark(s)</td><td>MA.8.AR.4.1</td></tr>
<tr><td colspan="2">$-7 = -\frac{9}{4}(4) + 2$
$-7 = -9 + 2$
$-7 = -7$

$-7 = 3(4) - 19$
$-7 = 12 - 19$
$-7 = -7$
☐ Yes</td><td>Recommended Scoring

(6 Total Points)</td><td>Students who only show their work for one of the equations and choose "yes" are showing a lack of understanding. Teachers should set the expectation of what constitutes as acceptable for work. Some students may have the ability to do calculations in their head or utilize the calculator well.</td></tr>
</table>

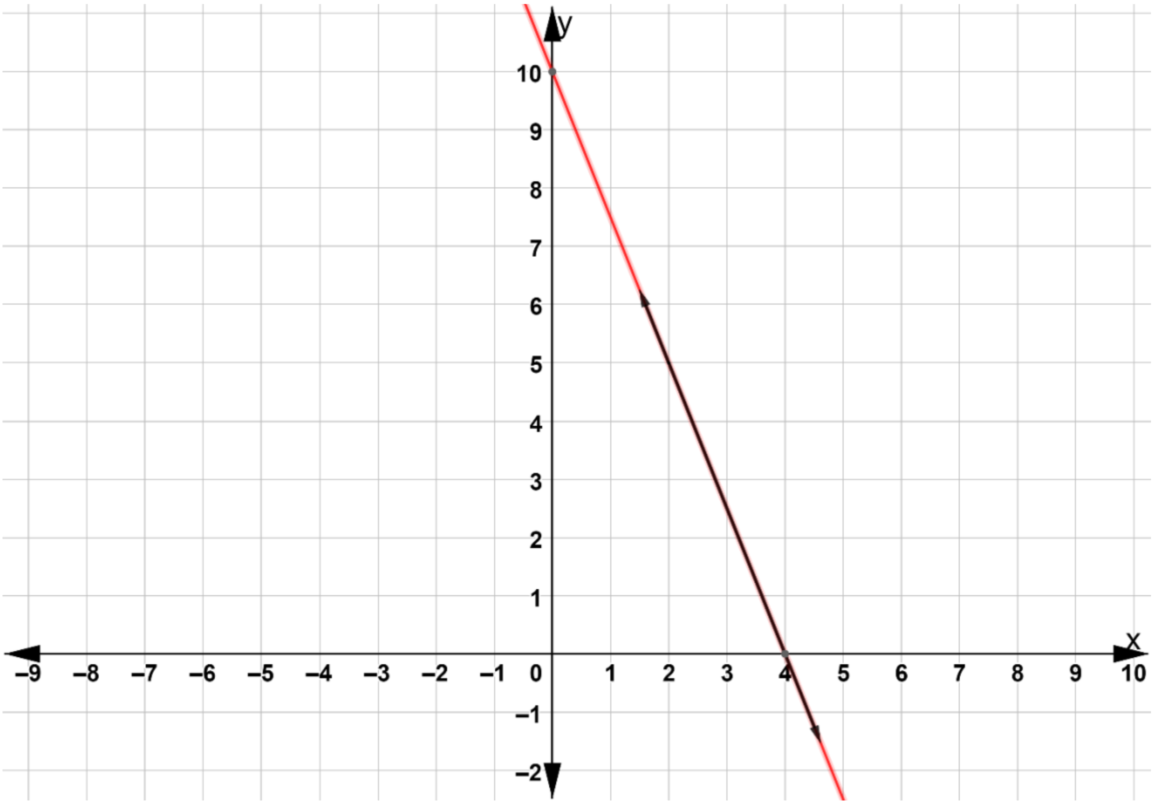
Question	7c	Benchmark(s)	MA.8.AR.4.1
$11 = -\frac{9}{4}(-4) + 2$ $11 = 9 + 2$ $11 = 11$ $11 = 3(-4) - 19$ $11 = -12 - 19$ $11 = -31$ ☐ No		**Recommended Scoring** **(6 Total Points)**	Students may only show their work for $y = -\frac{9}{4}x + 2$ because it is true for the point $(-4,11)$. Students who do this are not showing an understanding that the solution must be true for both equations. On the other hand, if students only show their work for $y = 3x - 19$ and choose "no," they are showing understanding.
Question	**8**	**Benchmark(s)**	MA.8.AR.4.1
The point (2,3) is [○ the slope / ○ a solution / ○ the y-intercept] of [○ $y = x + 1$ only. / ○ $y = -x + 5$ only. / ○ $y = -x + 5$ and $y = x + 1$.]		**Recommended Scoring** **(6 Total Points)**	This question addresses Clarification 1, "Instruction focuses on the understanding that a solution to a system of equations satisfies both linear equations simultaneously."

<table>
<tr><th>Question</th><th>9</th><th>Benchmark(s)</th><td>MA.8.AR.4.3</td></tr>
<tr><td colspan="2"><table><tr><th>Solution</th><td>$(-2, -6)$</td></tr></table></td><th>Recommended Scoring

(6 Total Points)</th><td>No partial credit is suggested.</td></tr>
<tr><th>Question</th><th>10</th><th>Benchmark(s)</th><td>MA.8.AR.4.3</td></tr>
<tr><td colspan="2">Answers may vary. The exact solution is $(-3.375, 5.75)$, but students are only expected to approximate from the graph. An example approximation students might give is shown. Reasonable approximations based on the graph should be accepted.
<table><tr><th>Approximate Solution</th><td>$(-3.5, 5.75)$</td></tr></table></td><th>Recommended Scoring

(8 Total Points)</th><td>Student solutions may vary but should be within a small range of the provided approximate solution.</td></tr>
<tr><th>Question</th><th>11</th><th>Benchmark(s)</th><td>MA.8.AR.4.3</td></tr>
<tr><td colspan="2"><table><tr><th>Approximate Solution</th><td>$(9, 147.15)$</td></tr></table></td><th>Recommended Scoring

(8 Total Points)</th><td>Students should be able to determine that the x-value is 9. Students with content knowledge should then substitute that value into one of the equations to determine the exact y-value. If students correctly estimate the y-value as being between 140 and 150 then consider deducting 2 points.</td></tr>
</table>

Question	12a	Benchmark(s)	MA.8.AR.4.3
		**Recommended Scoring** **(4 Total Points)**	No partial credit is suggested.

Question	12b	Benchmark(s)	MA.8.AR.4.3
The two equations in the system are lines with ○ the same slope. ○ different slopes. The system $\begin{cases} y = 10 - 2.5x \\ y = -\frac{5}{2}x + 10 \end{cases}$ ○ has no solution. ○ has one solution. ○ has an infinite number of solutions.		**Recommended Scoring** **(6 Total Points)**	Consider giving 3 points for the first answer box and 3 points for the second answer box.